COSC 3337 Data Science I Term Project Instructions

Goal The class project is a major component of your final grade. It is designed to give you hands-on experience applying course concepts to a real-world, creative problem that you can showcase on your resume or portfolio.

1. Group Structure & Grading Rules

- Group Size: Teams of **3 students** (formed by you. If you need help finding teammates, let me know.).
- Shared Grade: *All group members receive the same grade.* Choose your teammates carefully, establish regular communication, and divide computational and writing tasks evenly.
- Grading Focus: Grades are based on your learning process, execution, and honest reporting — *not on whether your model achieves high performance*. If an experiment fails or yields low accuracy, you will not be penalized, provided you analyze the issues clearly and suggest future improvements.
- The term project contributes **15%** to your final grade.
- Extra Credit: Team will receive **+2 pts** to your *final grade* if you participate in the Data Science Showcase on December 4. Team will receive **+4 pts** if you are selected for any of the awards at the Data Science Showcase. (e.g., if your *final grade* for this class is 88, then your grade will become 90 if you participate in the Data Science Showcase, 92 if your team wins any of the awards.) Must register for the showcase by Nov 1: https://hpedsi.uh.edu/partnerships/sponsored-events/data-science-showcase . For Mentor, please put in: Pam Qian.

2. Project Deliverables

There are two required deliverables and an optional showcase component.

Part 1: Project Proposal (Due Sunday Nov 1, 11:59 PM)

- Purpose: Get feedback from instructors/TAs on project feasibility and scope. *You are encouraged to attend Office Hours to discuss your project idea even if your idea is still vague.*

- Format: 1 page, max 2 pages, include references. Submit as a PDF. Figures are encouraged.
- Required Sections:
    1. Introduction and Motivation: What is the purpose of the project? Is it to detect cancer, study sterility, predict campus traffic..?: Why this project matters (societal impact, practical utility, or personal learning goals).
    2. Proposed Method: You will implement and compare at least two methods covered in this course: one traditional data science method (i.e., methods covered before Midterm 1) and one neural network. A small multilayer perceptron is sufficient; training a large model is not required. Explain why each method is appropriate for your task.
    3. Evaluation: Describe your training, validation, and test strategy and justify your evaluation metrics. Compare both methods using the same data partitions. Fit preprocessing steps using training data only, use validation data or cross-validation for model selection, and reserve the test set for final evaluation.
    4. Datasets: Find an existing dataset for your project. Report: dataset source and access link; approximate number of observations and features; and prediction target.
    5. Team Contributions: Clear breakdown of each member's specific coding, experimental, and writing responsibilities.
- Note: You may update your project approach later in the semester if you learn new methods in class.

Part 2: Final Project Report and Code Submission (Due Sunday Dec 6. 11:59 PM)

- Format: 6–8 pages, including references. Submit as a PDF document.
- Reproducibility Standard: The report must be detailed enough that a classmate could replicate your results without guesswork.
- Required Sections, include clearly labeled figures and tables that support your analysis:
    1. Introduction

2. Related Work
3. Proposed Method
4. Experiments
5. Results and Discussion
6. Conclusions
7. Team Contributions (Detailed breakdown of individual work)

- Code: Provide all your code in Python notebook(s) with all results and outputs, such as printed numbers, accuracy reports, validation results, testing results, and figures. Submit as .ipynb documents.

Part 3: Only For the Data Science Showcase Participants

You will need to submit a poster and slides later. Instructions can be found here under #submission: https://hpedsi.uh.edu/partnerships/sponsored-events/data-science-showcase

Grading Rubric

Component	Project points
Proposal	10
Problem definition, data understanding, and related work	15
Methods and implementation	25
Evaluation and experimental design	20
Results, interpretation, and limitations	20
Reproducibility and report presentation	10
Total	**100**